

Sustainability – what is it and how can you contribute?

<https://www.iop.org/lookingglass> - if you get a chance please listen to at least one of these podcasts



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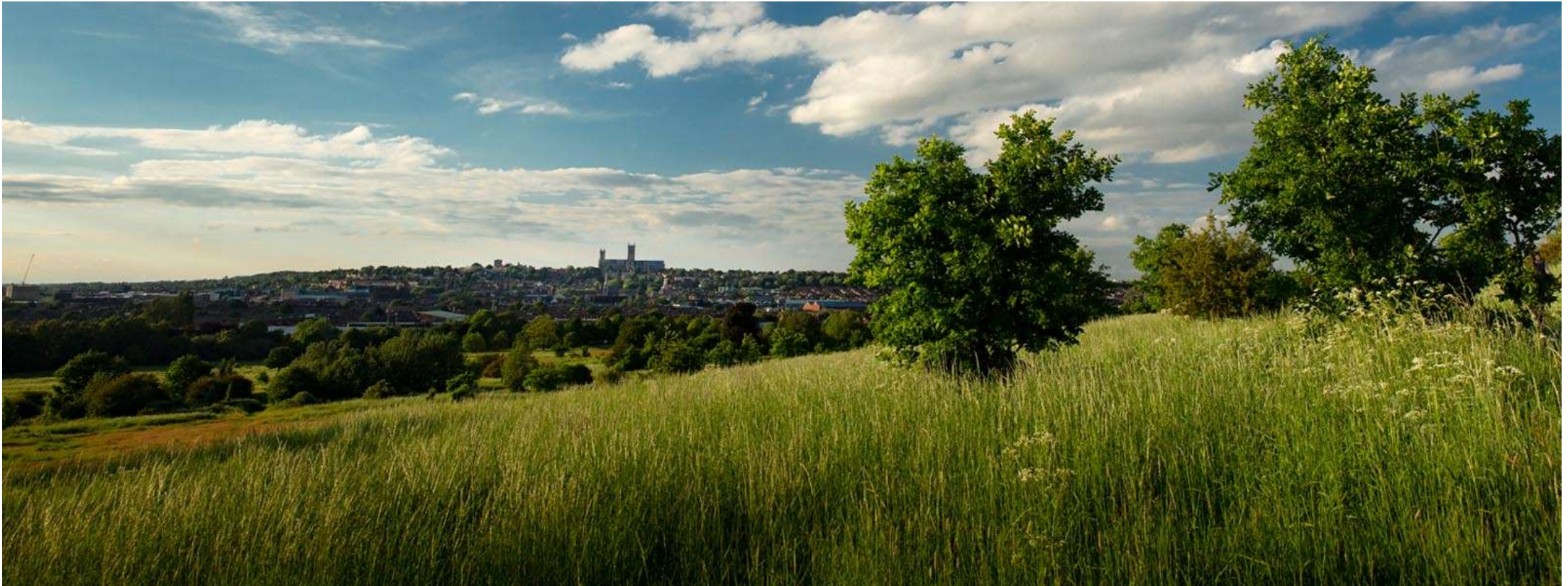
What does sustainability mean to you?

What words and ideas do you associate with sustainability?



Sustainability – an accepted definition?

“meeting the needs of the present without compromising the ability of future generations to meet their own needs.” UN Bruntland Commission 1987.

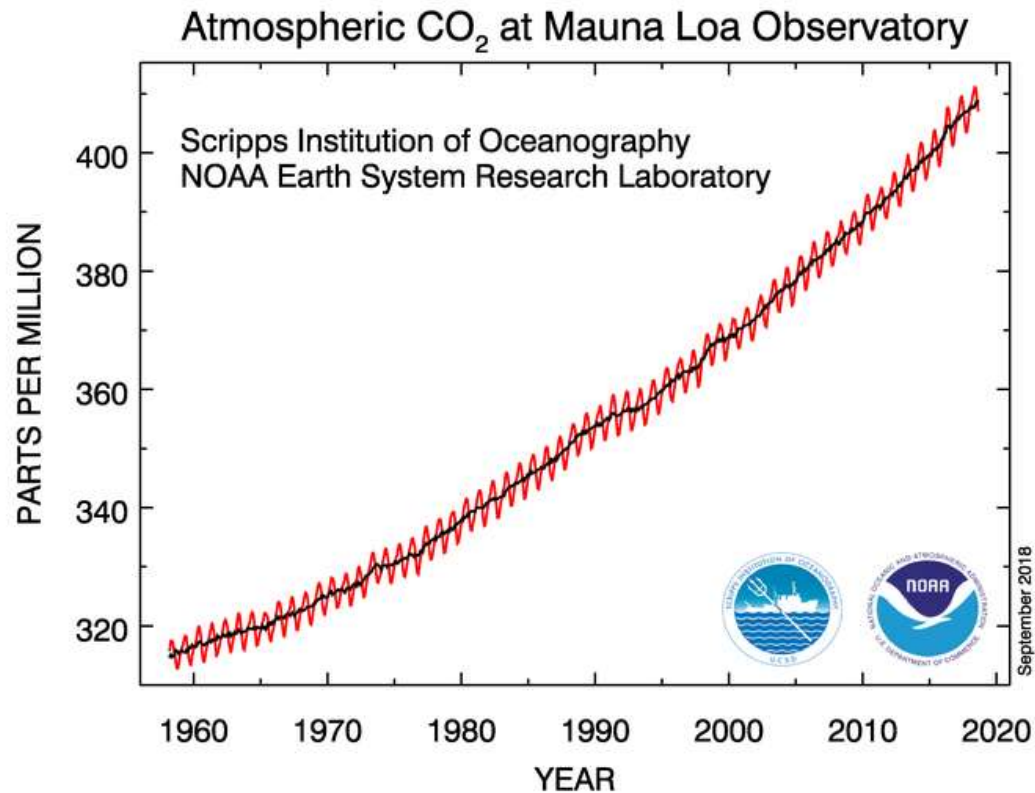


Meeting the needs of the present without compromising the ability of future generations to meet their own needs

- What questions does that raise for you?
- What is a need?
- Are we living in a way that will enable future generations to meet their needs?
- Evidence that we are or are not living sustainably?

The next few slides show a few of the things I care about that suggest we are not acting sustainably!

Climate change, driven by greenhouse gases, is already causing havoc. In addition to fires, floods and unpredictable harvests it is contributing to poor health outcomes in Europe and elsewhere. Is our current course sustainable?



Eunice Newton Foote, an American woman scientist, discovered that carbon dioxide was a greenhouse gas in 1856



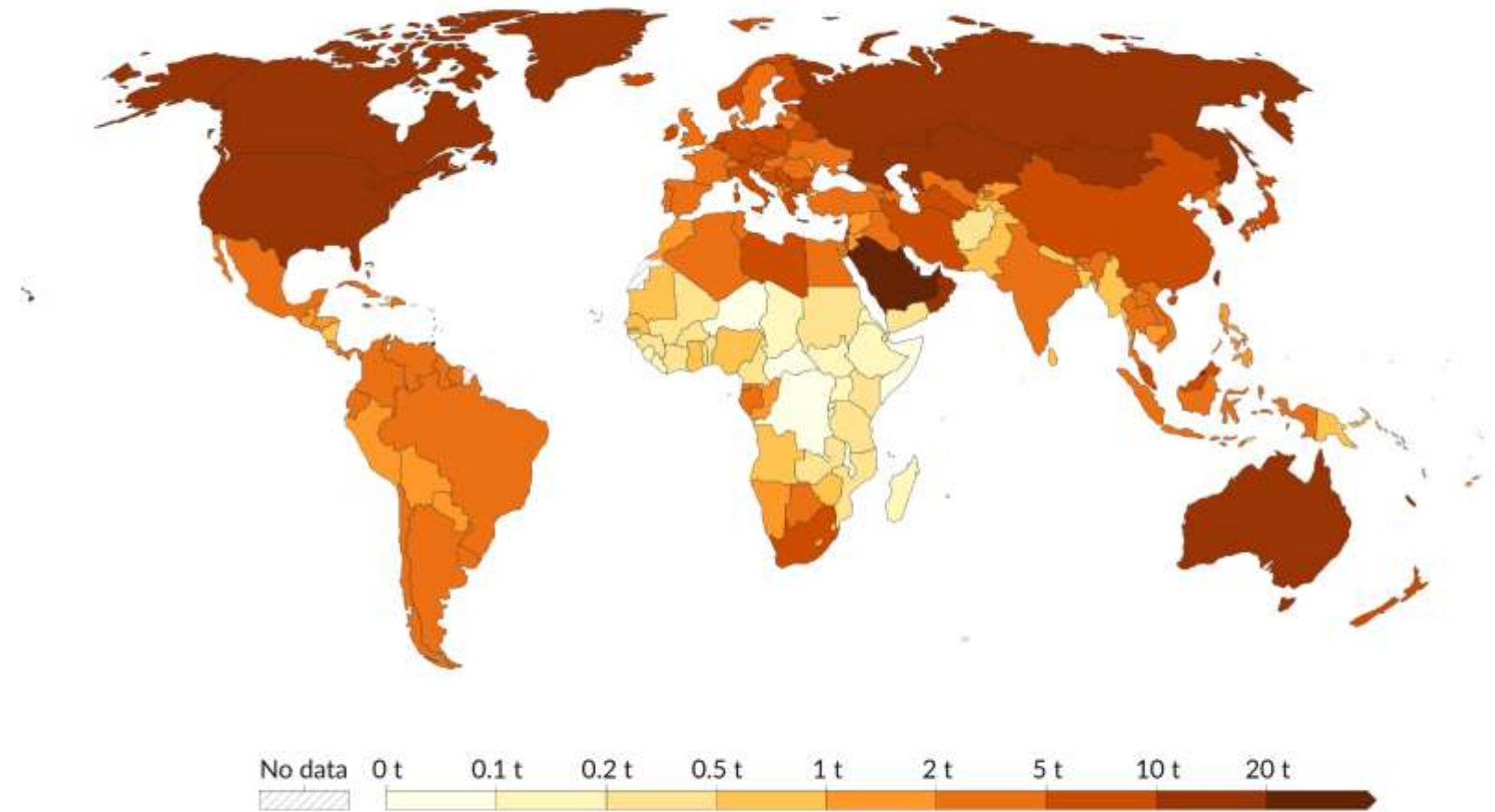
Tuvalo's Foreign Minister, Simon Kofe, delivers an address. His whole nation is threatened by rising sea levels.

Climate
change
largely
driven by
wealthier
nations
and the
wealthiest
people

Per capita CO₂ emissions, 2023

Carbon dioxide (CO₂) emissions from fossil fuels and industry¹. Land-use change is not included.

Our World
in Data



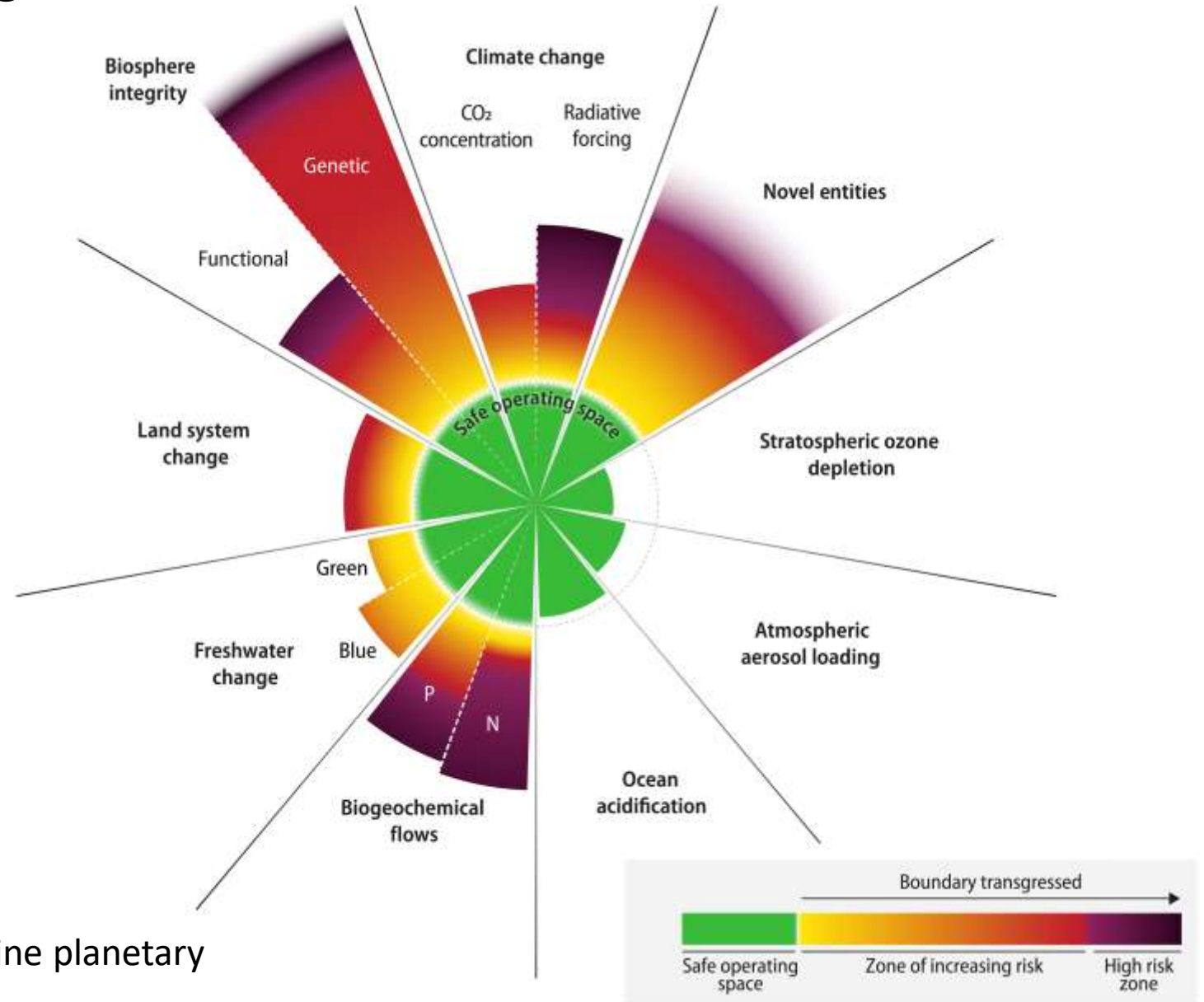
Data source: Global Carbon Budget (2024); Population based on various sources (2024)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

1. Fossil emissions: Fossil emissions measure the quantity of carbon dioxide (CO₂) emitted from the burning of fossil fuels, and directly from industrial processes such as cement and steel production. Fossil CO₂ includes emissions from coal, oil, gas, flaring, cement, steel, and other industrial processes. Fossil emissions do not include land use change, deforestation, soils, or vegetation.

Earth system scientists believe we have exceeded 6 of 9 planetary boundaries

- Which are critical for core life support functions?
- Which could you as future professionals influence?
- Should these boundaries be viewed separately as independent issues?

Richardson et al (2023) Earth beyond six of nine planetary boundaries Science Advances 9



The sixth mass extinction is probably underway

- Are we experiencing the Earth's 6th mass extinction event?
- Extinction rates are estimated to be 100-1000 times higher than background. For a quick read see what the scientists at the [Natural History Museum think](#).
- They think human activity is to blame
- See Cowie et al (2022) [The Sixth Mass Extinction: fact, fiction or speculation](#)? for more detail
- Mass extinctions 1-5 led to 75- 96% of species being lost (**including the dominant species**).



Many are hungry – many more food insecure

- [See WHO summary](#)
- Estimate 733 million hungry
- 2.3 billion are food-insecure
- 2.8 billion can't afford a healthy diet (including 71% of populations from low-income countries and 6.3% of people in high-income countries)
- Not on track to eliminate hunger by 2030

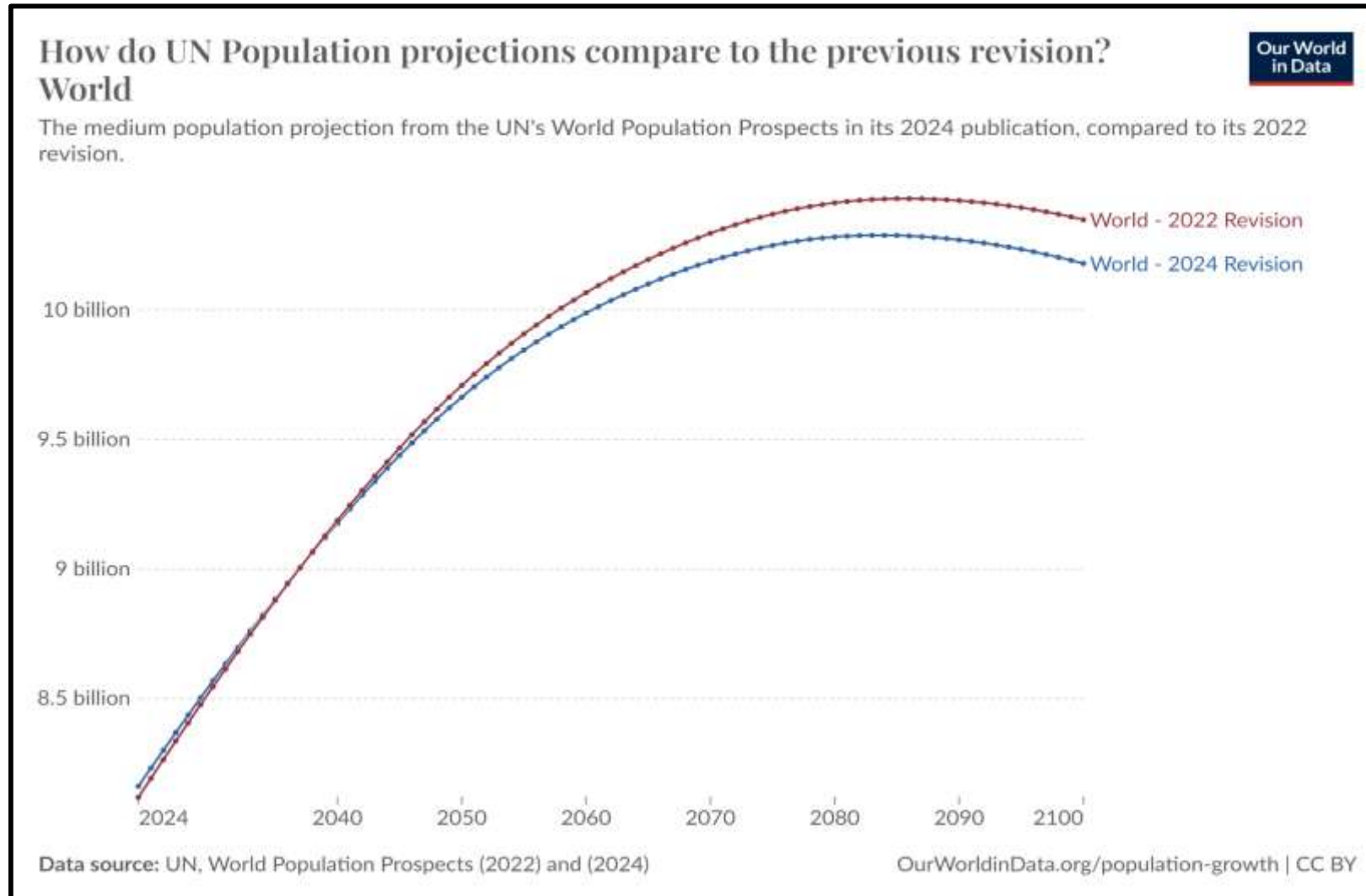


Any thoughts?

Sustainability is about people as well as the environment



Human population projections: peak mid 2080s but growth not evenly distributed across the globe



Sustainability is a multifaceted challenge and everyone can contribute to solutions

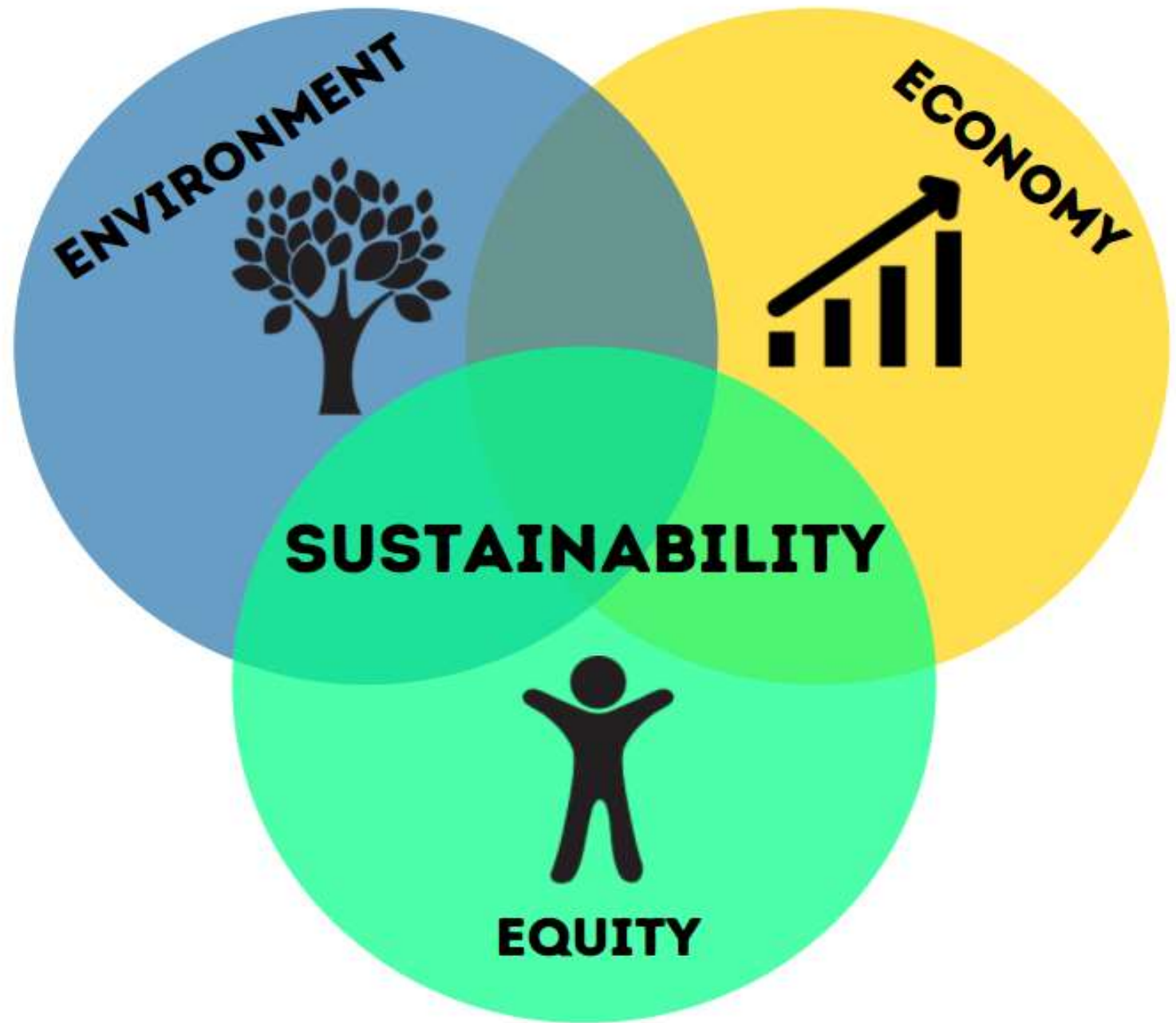


Image from:

<https://www.sustain.ucla.edu/wp-content/uploads/2020/10/3E2.png>

We have a global framework – the UN SDGs 2015 – 2030

You will find many online resources relevant to the SDGs – see for example <https://www.globalgoals.org/>



SDG	Goals
1 No Poverty	End poverty in all its forms everywhere
2 Zero Hunger	End hunger, achieve food security and improved nutrition, and promote sustainable agriculture
3 Good Health and Well-being	Ensure healthy lives and promote well-being for all at all ages
4 Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
5 Gender Equality	Achieve gender equality and empower women and girls
6 Clean Water and Sanitation	Ensure availability and sustainable management of water and sanitation for all
7 Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable and modern energy for all
8 Decent Work and Economic Growth	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
9 Industry, Innovation and Infrastructure	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

SDG	Goals
10 Reduced Inequality	Reduce inequality within and among countries
11 Sustainable Cities and Communities	Make cities and human settlements inclusive, safe, resilient and sustainable
12 Responsible Consumption and Production	Ensure sustainable consumption and production patterns
13 Climate Action	Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy
14 Life Below Water	Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15 Life on Land	Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification and halt and reverse land degradation and halt biodiversity loss
16 Peace, Justice and Strong Institutions	Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17 Partnerships for the Goals	Strengthen the means of implementation and revitalize the global partnership for sustainable development

SDG framework

- Each SDG has targets and indicators (see example of first 3 targets below)
- Some of the indicators are used for more than one target
- Some are hard to measure
- A list of targets and indicators is [here](#)

<i>Goals and targets (from the 2030 Agenda)</i>	<i>Indicators</i>
Goal 1. End poverty in all its forms everywhere	
1.1 By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day	1.1.1 Proportion of population below the international poverty line, by sex, age, employment status and geographical location (urban/rural)
1.2 By 2030, reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions	1.2.1 Proportion of population living below the national poverty line, by sex and age 1.2.2 Proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions
1.3 Implement nationally appropriate social protection systems and measures for all, including floors, and by 2030 achieve substantial coverage of the poor and the vulnerable	1.3.1 Proportion of population covered by social protection floors/systems, by sex, distinguishing children, unemployed persons, older persons, persons with disabilities, pregnant women, newborns, work-injury victims and the poor and the vulnerable

Which of the UN SDGs will you contribute to?

- Pick the SDG that matters most to you
- Which SDGs support it?
- Which SDGs depend on it?
- How is it doing?
- How will your work help to progress it?
- Which SDGs do not need the expertise of someone with a physics or mathematics degree?
- What (non technical) skills do you think you need to contribute to sustainable development?



Contributing to Sustainable Development

- UNESCO key competencies for SD
- How many do you have?

Key competencies for sustainability

WAYS OF THINKING

- KC1 Systems thinking competency:** the abilities to recognise and understand relationships; to analyse complex systems; to think of how systems are embedded within different domains and different scales; and to deal with uncertainty.
- KC2 Anticipatory competency (future thinking):** the abilities to understand and evaluate multiple futures – possible, probable and desirable; to create one's own visions for the future; to apply the precautionary principle; to assess the consequences of actions; and to deal with risks and changes.
- KC3 Critical thinking competency:** the abilities to question norms, practices, and opinions; to reflect on one's own values, perceptions, and actions; and to take a position in the sustainability discourse.

WAYS OF PRACTICING

- KC4 Strategic competency:** the abilities to collectively develop and implement innovative actions that further sustainability at the local level and further afield.

- KC5 Collaboration competency:** the abilities to learn from others; to understand and respect the needs, perspectives and actions of others (empathy); to understand, relate to and be sensitive to others (empathic leadership); to deal with conflicts in a group; and to facilitate collaborative and participatory problem solving.

- KC6 Integrated problem-solving competency:** the abilities to apply different problem-solving frameworks to complex sustainability problems and develop viable, inclusive and equitable solution options that promote sustainable development, integrating the above-mentioned competences.

WAYS OF BEING

- KC7 Self-awareness competency:** the abilities to reflect on one's own role in the local community and (global) society; to continually evaluate and further motivate one's actions; and to deal with one's feelings and desires.
- KC8 Normative competency:** the abilities to understand and reflect on the norms and values that underlie one's actions; and to negotiate sustainability values, principles, goals, and targets, in a context of conflicts of interests and trade-offs, uncertain knowledge and contradictions.

Source: UNESCO's key competencies for sustainability (UNESCO, 2017) - ESD guidelines published in the UK by QAA and Advance HE (2021)

The SDGs are not on track to deliver key outcomes by 2030

- Derailed by global events
- But could they have ever delivered by 2030?
- They are contested
- But valuable in setting united ambition?
- What do you think?

CURRENT STATE OF PROGRESS TOWARD THE SUSTAINABLE DEVELOPMENT GOALS
BASED ON SELECT TARGETS



¹ Distance from target (2023) and trend of Sustainable Development Goals progress (2023) refer to current level and trend information for the latest available data utilizing the calculation methodology

The goals are widely used in business e.g.:



The United Nations Sustainable Development Goals guide reflection and action on the most pressing challenges and opportunities facing humanity and the natural world. Whilst we will deliver against a number of the goals we have selected five where Microngreen will deliver the greatest benefit.



OUR CLIMATE

ECONOMY

INDUSTRY, INNOVATION AND INFRASTRUCTURE

New Campus to be carbon neutral and flagship for sustainability with the application of the 3Is at all our sites including Micronclean India.

BIOSPHERE

CLIMATE ACTION

Setting our pathway to zero for our carbon emissions per unit and supply chain emissions with a 50% reduction of our direct emissions by the end of 2027.



CREATING OUR CIRCULAR ECONOMY

ECONOMY

RESPONSIBLE CONSUMPTION AND PRODUCTION

Build our circular economy; reducing water consumption, reusing and recycling packaging, eliminating waste to landfill.



OUR PEOPLE & COMMUNITIES

ECONOMY

GOOD HEALTH AND WELL-BEING

Commitment to healthy lives and promotion of wellbeing for all at all levels of the business through policy and empowerment of our people.

BIOSPHERE

PARTNERSHIPS FOR THE GOALS

Working in partnership to deliver initiatives to bring positive impact to our local communities and to deliver against SDGs 3, 9, 12 and 13.



SUSTAINABLE DEVELOPMENT GOALS

Local employer examples

- [Siemens Energy](#) sustainability report says they focus on SDGs 5, 7, 8, 9 and 13
- Bakkavor ESG (Environment Social and Governance) strategy [refers to the SDGs](#) and says they make a significant contribution to 3, 5, 8, 12 and 13.
- United Lincolnshire Hospitals Trust [Green Plan](#) refers to the SDGs and says they are working to align them to their Green Plan
- Enterprise (car rentals) – [ESG report](#) says aligned to SDGs 6,8,9,10 and 12
- We can evaluate these contributions tomorrow

Calculate your own carbon footprint

<https://footprint.wwf.org.uk/>



What is your university doing?

Sustainability is embedded in our strategy

<https://www.lincoln.ac.uk/strategicplan/environmentalsustainability/>



Some local sustainability organisations

- Lincoln Climate Commission <https://www.lincolnclimate.org.uk/>
- Lincoln Conservation Group
<http://lincolnconservationgroup.weebly.com/about-us.html>
- Lincolnshire Wildlife Trust <https://www.lincstrust.org.uk/>
- Local motion <https://localmotion.org.uk/>
- EarthLincs – Lincoln's climate emergency centre
<https://www.facebook.com/groups/181394503851074/>
- Mint Lane Café <https://www.mintlanecafe.com/>
- Green Synergy <https://greensynergy.org.uk/>

Sustainability 'myths' - which did you/do you believe?

- Read - <https://www.scientificamerican.com/article/top-10-myths-about-sustainability/>
- **Nobody knows what sustainability really means.**
- **Sustainability is all about the environment.**
- **“Sustainable” is a synonym for “green.”**
- **It’s all about recycling.**
- **Sustainability is too expensive.**
- **Sustainability means lowering our standard of living.**
- **Consumer choices and grassroots activism, not government intervention, offer the fastest, most efficient routes to sustainability.**
- **New technology is always the answer.**
- **Sustainability is ultimately a population problem.**
- **Once you understand the concept, living sustainably is a breeze to figure out.**

- Forvis Mazars - senior accountant: <https://uk.linkedin.com/in/madeleine-long-80139518b>. I mentioned this company as I know they do sustainability consultancy.
- Boots - data analyst. This graduate has now moved on to Brightwell: <https://uk.linkedin.com/in/lauren-coupe>
- National Physical Laboratory - graduate research scientist in quantum technology: <https://www.linkedin.com/in/jodie-renaud-22564a233/>
- Rolls-Royce - transformation associate: <https://www.linkedin.com/in/leah-ward1/>
- BT - operations manager: <https://www.linkedin.com/in/niall-garry-867949ba/>
- VTT - research scientist: <https://www.linkedin.com/in/jake-muff/>
- Juniper Networks - virtual partner account manager: <https://www.linkedin.com/in/felicity-levett/>
- Leonardo - project engineering lead, previously worked as environmental engineering team lead at Teledyne e2v: <https://www.linkedin.com/in/robert-sharp-110a34143/>
- Baker Perkins - electrical design engineer: <https://www.linkedin.com/in/anthony-w-54882818b/>

National and International Businesses that have employed our graduates

- BT: <https://business.bt.com/about-us/sustainability/#:~:text=Net%20zero%20by%202031&text=Today%20C%20we%20use%20100%25%20renewable,and%20customer%20emissions%20by%202041>.
- BOOTS: <https://www.boots-uk.com/environmental-social-governance/what-we-do/progressing-our-sustainable-bootprint/>
- RR: <https://www.rolls-royce.com/sustainability.aspx>
- Forvis Mazars: <https://www.forvismazars.com/group/en/who-we-are/news-publications/sustainability-reports>